



Assignment submission guidelines

- Submit assignment on Teams. Submit all theory answers in a PDF document with proper numbered questions and page numbers.
- For simulation questions, include plots and inferences in the PDF and submit the code separately in proper format (.m, .py, etc).
- Only two files should be submitted: one PDF and one code file. Do not submit more than two files.
- Do not zip or compress the files.
- Give detailed steps and justification required to obtain the solution.
- There will be negative marking for late submissions.
- Academic integrity is essential. Copying of assignments is intolerable and, if found guilty, serious action will be taken.

Q1. A simplified m longitudinal motion of an underwater vehicle can be described by

$$m\ddot{x} + k\dot{x}|\dot{x}| = u \quad (1)$$

where x is the position of the vehicle, u is the control input (a force that is provided by a propeller), m is the mass of the vehicle, and $k > 0$ is the drag coefficient. Assuming the value of m is known exactly, the drag coefficient is bounded $k_1 \leq k \leq k_2$ and the position and its derivative (velocity) $x, \dot{x}$ are measured:

- (a) Obtain a state system model of the vehicle using $x_1 = x, x_2 = \dot{x}$ as the state variables.
- (b) Design a conventional sliding mode control law u that drives $x_1, x_2 \rightarrow 0$ as time increases.
- (c) Simulate the control system for $x_1(0) = 2$ m, $x_2(0) = 0.5$ m/s, $m = 4$ kg, and $k = 1.5 + 0.4\sin(2t)$ [kg/ms]. Plot the time histories of the sliding variable, the control function u , the position x_1 , and the velocity x_2 .
- (d) Identify the quantities that reach zero in finite time and the ones that approach zero asymptotically.

Hint: The function $k\dot{x}|\dot{x}|$ can be bounded as $|k\dot{x}|\dot{x}|| \leq k_2\dot{x}^2 = 1.9\dot{x}^2$. **(2+6+10+2=20)**

Q2. Repeat part (c) of the above problem by approximating the sign function in the control law by the sigmoid function $\text{sign}(s) \approx \frac{s}{|s| + \epsilon}$ and separately by the saturation function given below for some $\epsilon > 0$

$$\text{sign}(s) \approx \text{sat}(s, \epsilon) = \begin{cases} 1 & \text{if } s > \epsilon \\ \frac{s}{\epsilon} & \text{if } |s| \leq \epsilon \\ -1 & \text{if } s < -\epsilon \end{cases} \quad (2)$$

Q3. Consider the following system

(2+3+4+12+11=32)

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix} x + \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} (u + d), \quad (3)$$

where $x \in \mathbb{R}^3$, $u \in \mathbb{R}^2$, $d \in \mathbb{R}^2$ with disturbance $d = [2 \sin(t) + \cos(t) \quad 1 + \cos(2t)]^\top$.

- (a) Find the equilibrium point of the above system for $d = 0$.
- (b) Design the following controllers for stabilizing the equilibrium point of the above system:
 1. Continuous state feedback controller.
 2. Sliding mode controller.
 3. Replace $\text{sign}(s)$ in sliding mode control with saturation function $\text{sat}(s, \epsilon)$ for $\epsilon = 0.01$. Compare the results of the simulations.
 4. Replace $\text{sign}(s)$ in sliding mode control with approximation $\frac{s}{|s| + \epsilon}$.

One can use different values of ϵ in parts (3) and (4). Note that the definitions of both approximations are component-wise.

Show the steps involved in the design procedure and simulate the designed controllers. Take initial condition $x(0) = [2 \quad 5 \quad 3]^\top$. Plot all states of the system, control inputs, and sliding variables. [Note: 12 points for codes of different controllers and 11 points for plots.]

Q4. Design a non-singular terminal sliding mode control for the system: (4+4+4+4+4=20)

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= f(x) + g(x)(u + d) \end{aligned}$$

where $x_1, x_2 \in \mathbb{R}$, $u \in \mathbb{R}$, $d \in \mathbb{R}$ and disturbance $d = 1.5 + 0.5 \cos t$. The functions $f(x)$ and $g(x)$ are known functions. Prove the stability of the closed-loop system and calculate the time of convergence of the sliding variable and the states of the system. Perform simulation for $f(x) = x_1^2 + 2x_2$, $g(x) = 2$ and initial conditions $x_1(0) = 2$ and $x_2(0) = 3.5$. Plot states, sliding variable, control input, and phase portrait.